

Statistical modeling

Module 09

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Binary outcome

Linear regressions are models for analyzing the links between a set of variables x for a **numerical** outcome y . What can be done when y is not numerical but categorical and more specifically binary?

Example: In a marketing study, one wants to detect if customers are likely to renew their subscription for an online sport information service.

The data set: 60 users, response is $y = 1/0$ (renewed/not renewed), and x 's are

- ▶ Age: age category,
- ▶ AvPages.Week: Avg # of pages viewed per week,
- ▶ Pages.Week: # of pages viewed during the last (previous) week.

Source: [https://help.xlstat.com/customer/en/portal/articles/](https://help.xlstat.com/customer/en/portal/articles/2062235-running-a-logistic-regression-with-xlstat?b_id=9283)

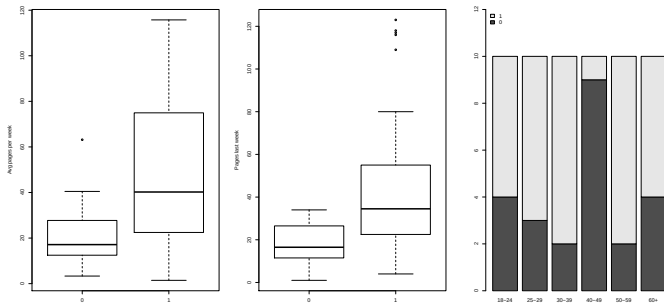
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Binary outcome

Exploratory analysis:

- ▶ To inspect the relation between y and a numerical x , represent x per level of y ; for example make one boxplot of x for $y = 0$ and one for $y = 1$.
- ▶ To inspect the relation between y and a categorical x , represent a table of frequency of y per level of x ; for example, one bar plot of frequencies of $y = 0/1$ per category of x .

Binary outcome



Renewal ($y = 0/1$) is

- ▶ positively associated with high avg # pages per week (left plot).
- ▶ positively associated with high # pages viewed last week (middle plot).
- ▶ negatively associated with age category 40 – 49 (right plot).

Binary outcome

Exploratory data analysis with a binary outcome has the same limit as with a numerical outcome, namely, that the associations which have been observed are not conditional on the other ones but overall. There may be collinearities here.

Logistic regression is a regression method that allows to estimate/test links between the variables in the same way as regression does for numerical variables.

Logistic regression: concept

Suppose that we can observe a population of kids. We are interested into knowing if they like chocolate or not ($y = 1/0$), according to the sex Girl and Boy $x = B/G$. The result of the analysis would be the proportions per category. For example,

	Boy	Girl
$Y = 1$	$P(Y = 1 B) = 0.65$	$P(Y = 1 G) = 0.53$
$Y = 0$	$P(Y = 0 B) = 0.35$	$P(Y = 0 G) = 0.47$

Of course, in general, we cannot observe the whole population and cannot know these numbers. We need to draw a sample at random.

Logistic regression: concept

Drawing an observation at random (e.g. selecting a kid and asking him whether he/she likes chocolate), the response Y can be 1 or 0.

The probability depends on the variable x (e.g. the sex of the kid). That is, the response Y are distributed according to a Bernoulli distribution with parameter $p(x)$, where x are the independent variables:

$$P(Y = 1|x) = p(x), \quad P(Y = 0|x) = 1 - p(x)$$

In the example,

$$P(Y = 1|B) = p(B) = 0.65, \quad P(Y = 1|G) = p(G) = 0.53.$$

$$P(Y = 0|B) = 1 - p(B) = 0.35, \quad P(Y = 0|G) = 1 - p(G) = 0.47.$$

Logistic regression: linear predictor

With logistic regression, the probability $p(x)$ is linked with x through the linear predictor $\eta(x)$

$$\eta(x) = b_0 + b_1x_1 + \cdots + b_px_p.$$

In particular, the predicted probability given x is

$$p(x) = \frac{\exp\{\eta(x)\}}{1 + \exp\{\eta(x)\}}$$

The link function, transforming $p(x)$ in $\eta(x)$ is called the logit function

$$\eta(x) = \log \left(\frac{p(x)}{1 - p(x)} \right).$$

Logistic regression: example

Example: For the kids example,

$$\eta(B) = \log \left(\frac{0.65}{1 - 0.65} \right) = 0.62.$$

and

$$\eta(G) = \log \left(\frac{0.53}{1 - 0.53} \right) = 0.12.$$

Using a dummy variable notation with G as the reference, we can write

$$\eta(x) = b_0 + b_1 \delta_B(x) = 0.12 + 0.5 \times \delta_B(x).$$

Logistic regression: interpretation

Coefficient $b_1 = 0.5$ quantifies the difference between girls and boys in terms of the probability that $Y = 1$. It plays the role of a contrast like in regression. In particular,

- ▶ if $b_1 = 0$, there would be no difference between boys and girls.
- ▶ if $b_1 > 0$, then $p(B) > p(G)$
- ▶ if $b_1 < 0$, then $p(B) < p(G)$

Logistic regression: prediction

For a new x , the prediction $\hat{y}(x)$ is obtained by predicting the linear predictor, the probability, and then the outcome:

1. The linear predictor

$$\hat{\eta}(x) = \hat{b}_0 + \hat{b}_1 x_1 + \cdots + \hat{b}_p x_p$$

2. The probability

$$\hat{p}(x) = \frac{\exp\{\hat{\eta}(x)\}}{1 + \exp\{\hat{\eta}(x)\}}$$

3. The outcome

$$\hat{y}(x) = \begin{cases} 0, & \text{if } \hat{p}(x) < 0.5 \\ 1, & \text{if } \hat{p}(x) \geq 0.5 \end{cases}$$

Logistic regression: example

Example: with the marketing study example the estimate is

```
> renew.mod <- glm(Renewed~Pages.Week + AvPages.Week + Age, data=Renew,
  family="binomial")
> summary(renew.mod)
[...]
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.35667	1.24579	-1.892	0.0585 .
Pages.Week	0.08928	0.03564	2.505	0.0123 *
AvPages.Week	0.02350	0.02585	0.909	0.3634
Age25-29	0.68781	1.20624	0.570	0.5685
Age30-39	0.96324	1.27573	0.755	0.4502
Age40-49	-2.98274	1.37732	-2.166	0.0303 *
Age50-59	1.08553	1.16754	0.930	0.3525
Age60+	0.30892	1.26437	0.244	0.8070

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

[...]

Logistic regression: prediction

The fitted linear predictor is

$$\begin{aligned}\hat{\eta}(x) = & -2.4 + 0.09x_1 + 0.02x_2 + 0.69\delta_{g_1}(x_3) \\ & + 0.96\delta_{g_2}(x_3) - 2.98\delta_{g_3}(x_3) + 1.09\delta_{g_4}(x_3) \\ & + 0.31\delta_{g_5}(x_3)\end{aligned}$$

where

- ▶ x_1 is "pages last week",
- ▶ x_2 is "Avg pages per week",
- ▶ x_3 is "Age" with levels $g_0 = "18 - 24"$ (the reference), $g_1 = "25 - 29"$, $\dots$, $g_5 = "60 +"$.

Logistic regression: prediction

For a customer with characteristics $x_1 = 30$, $x_2 = 25$, $x_3 = "18 - 24"$, the prediction is thus

$$\hat{\eta}(x) = -2.4 + 0.09 \times 30 + 0.02 \times 25 = 0.91,$$

$$\hat{p}(x) = e^{0.91} / (1 + e^{0.91}) = 0.71$$

$$\hat{y}(x) = 1$$

If the customer is in age category $x_3 = "40 - 49"$, then

$$\hat{\eta}(x) = -2.4 + 0.09 \times 30 + 0.02 \times 25 - 2.98 = -2.07,$$

$$\hat{p}(x) = e^{-2.07} / (1 + e^{-2.07}) = 0.11$$

$$\hat{y}(x) = 0$$

These examples show us how the probability varies when switching level for the categorical variable.

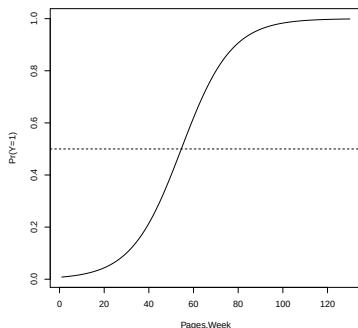
Logistic regression: interpretation

In the above example, the model has estimated that

- ▶ Within the population of customers with $x_1 = 30$, $x_2 = 25$, $x_3 = "18 - 24"$, the proportion of renewing customers is 71% (and 29% not renewing),
- ▶ Within the population of customers with $x_1 = 30$, $x_2 = 25$, $x_3 = "40 - 49"$, the proportion of renewing customers is 11% (and 89% not renewing).

Logistic regression: interpretation

Furthermore, let's consider the sub-population of customers with $x_2 = 20$, $x_3 = "40 - 49"$. The proportions of renewing customers ($y = 1$) in function of x_1 is estimated by the model as



The dashed line at 0.5 is the limit at which the prediction switch from $\hat{y}(x) = 0$ (below) to $\hat{y}(x) = 1$ (above).

Logistic regression: interaction

Interaction effects can be included in the model and are interpreted exactly the same way as in regression: "how the level of one variable moderates the effect of another one"

Example: with the previous model, the effect of `Pages.Week` was estimated to 0.08928. The researchers suspect that this effect varies according to the age. They include an interaction term between `Age` and `Pages.Week` in the logistic regression.

Logistic regression: interaction

```
> renew.mod2 <- glm(Renewed~Pages.Week*Age + AvPages.Week, data=Renew,  
                    family="binomial")
```

```
> summary(renew.mod2)
```

```
[...]
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-6.475e-01	1.337e+00	-0.484	0.628
Pages.Week	2.470e-02	4.430e-02	0.557	0.577
Age25-29	-1.530e+00	2.218e+00	-0.690	0.490
Age30-39	-6.538e-01	2.041e+00	-0.320	0.749
Age40-49	-1.334e+02	1.894e+04	-0.007	0.994
Age50-59	2.277e-01	2.434e+00	0.094	0.925
Age60+	-2.022e+00	2.187e+00	-0.925	0.355
AvPages.Week	1.486e-02	2.517e-02	0.590	0.555
Pages.Week:Age25-29	1.223e-01	1.221e-01	1.001	0.317
Pages.Week:Age30-39	7.373e-02	8.744e-02	0.843	0.399
Pages.Week:Age40-49	3.412e+00	5.038e+02	0.007	0.995
Pages.Week:Age50-59	3.012e-02	1.005e-01	0.300	0.764
Pages.Week:Age60+	1.323e-01	1.441e-01	0.918	0.359

```
[...]
```

Logistic regression: interaction

Because Age has lots of levels, the model has now a lot of coefficients. The interpretation remains the same as for a regression. Considering the slope in front of variable `Pages.Week` (x_1), then

- ▶ If Age is "18-24" then it is $0.02470x_1$
- ▶ If Age is "25-29" then it is $(0.02470 - 0.1223)x_1$
- ▶ If Age is "30-39" then it is $(0.0246 - 0.07373)x_1$
- ▶ etc.
- ▶ If Age is "60+" then it is $(0.0246 - 0.1323)x_1$.

Question: looking at the coefficients, in which age category is the slope of x_1 the highest? The lowest?

Logistic regression: prediction "interval"

For logistic regression, there is no equivalent to the prediction interval of the regression. It is rather obtained by giving the estimation of the probability $\hat{p}(x)$.

Examples: Suppose that the predictions of a customer are

1. $\hat{y} = 1$ with $\hat{p} = 0.99$: the prediction is quite certain.
2. $\hat{y} = 1$ with $\hat{p} = 0.53$: the prediction is quite uncertain.

In the case of regression, with the same prediction $\hat{y}$,

- ▶ Case $\hat{p} = 0.99$ would correspond to a narrow prediction interval.
- ▶ Case $\hat{p} = 0.53$ would correspond to a wide prediction interval.

Logistic regression: estimation

For a regression, the estimate $\hat{b}$ is the OLS.

For logistic regression, it is the maximum likelihood estimator (MLE).

This principle is to get $\hat{b}$ such that the model gives the maximum probability of observing what was actually observed. Consider the simple example below.

Logistic regression: estimation

With $n = 2$ (two observations), assume that the observed sample is $y_1 = 0$, $y_2 = 1$ with x_1 and x_2 given. Under the model (Bernoulli), the probability of observing this sample is

$$\begin{aligned}P(\text{"observe this sample"}) &= P(y_1 = 0, y_2 = 1) \\&= P(y_1 = 0)P(y_2 = 1) \\&= \{1 - p(x_1)\}p(x_2) \\&= \frac{1}{1 + \exp\{\eta(x_1)\}} \frac{\exp\{\eta(x_2)\}}{1 + \exp\{\eta(x_2)\}}\end{aligned}$$

Logistic regression: MLE

The maximum likelihood principle says that one should select $\hat{b}$ which maximizes this quantity, which is called the likelihood.

That is

$$\hat{b} = \arg \max_b L(b),$$

where

$$L(b) = \frac{\exp\{\eta(x_2)\}}{(1 + \exp\{\eta(x_1)\})(1 + \exp\{\eta(x_2)\})},$$

and

$$\eta(x) = b_0 + b_1x_1 + \cdots + b_px_p.$$

The estimator $\hat{b}$ hence obtained is called the maximum likelihood estimator (MLE).

Logistic regression: MLE

In practice, to obtain the MLE, it is easier and equivalent to maximize the logarithm of the likelihood, so called the log-likelihood:

$$\hat{b} = \arg \max_b \ell(b),$$

where

$$\ell(b) = \eta(x_2) - \log \left\{ 1 + e^{\eta(x_1)} \right\} - \log \left\{ 1 + e^{\eta(x_2)} \right\}$$

Note: The maximum likelihood is a general principle that can be applied to lots of model. E.g. when applied to linear regression, the OLS is obtained.

Logistic regression: MLE

In general, for a sample of size n :

$$\hat{b} = \arg \max_b \sum_{i=1}^n y_i \eta(x_i) - \sum_{i=1}^n \log \left\{ 1 + e^{\eta(x_i)} \right\}$$

where

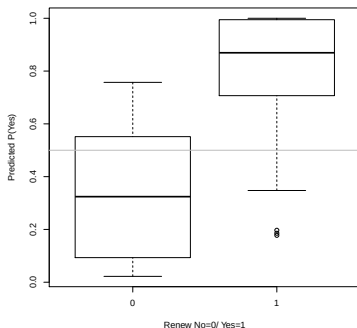
$$\eta(x_i) = b_0 + b_1 x_{i1} + \cdots + b_p x_{ip}.$$

To solve this, one uses numerical algorithms like Fisher scoring and Iterative Re-weighted Least Squares. The practical consequence is that one needs mathematical or statistical computer programs to use logistic regression.

Logistic regression: inspection of the prediction quality

For linear regression, the graphical way to inspect the quality of the prediction is the plot of the observed value y against the fitted values $\hat{y}$.

For logistic regression this is replaced by two boxplots of $\hat{p}(x_i)$, one for $y_i = 0$ and one for $y_i = 1$.



Logistic regression: inspection of the prediction quality

- ▶ On the y -axis, the fitted probability, on the x -axis, the observed class.
- ▶ The misclassified observations are above 0.5 for $y = 0$ and below 0.5 for $y = 1$.

The more the two boxplots are separated away from 0.5, the better the model.

Indeed, if the boxplots are very well separated, then it means that the fitted probabilities are either large (close to 1) or low (close to 0) and thus the model predicts the class precisely.

Logistic regression: inspection of the prediction quality

Unlike for linear regression, there is no R^2 's statistics for logistic regression.

A numerical way of inspecting the model quality is to build the so-called confusion matrix. Below

- ▶ TP stands for "True Positive" and FP for "False Positive",
- ▶ TN stands for "True Negative" and FN for "False Negative".

	Pred. 1	Pred. 0
Obs.1	TP	FN
Obs. 0	FP	TN

Logistic regression: inspection of the prediction quality

Example: with marketing data, "1" means "Renewed".

	Pred. 1	Pred. 0	Total
Obs.1	30 (TP)	6 (FN)	36
Obs. 0	7 (FP)	17 (TN)	24
Total	37	23	60

The accuracy is the proportion of correct prediction. It's a score for the model quality:

$$A = \frac{30 + 17}{60} = 78.3\%$$

Logistic regression: inspection of the prediction quality

Accuracy is a rough measure. The two following conditional probabilities are often inspected:

- ▶ Sensitivity: probability of predicting 1 given it is indeed 1

$$\frac{TP}{TP + FN} = P(\text{"Pred. 1"} | \text{"Obs. 1"}) = \frac{30}{36}$$

- ▶ Specificity: probability of predicting 0 given it is indeed 0

$$\frac{TN}{TN + FP} = P(\text{"Pred. 0"} | \text{"Obs. 0"}) = \frac{17}{24}$$

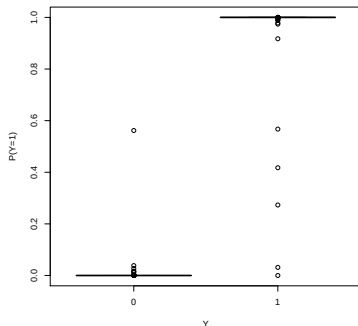
A good model has a large sensitivity and a large specificity¹.

¹Please don't ask how much is large...

Logistic regression: ROC curve

A finer inspection tool of the logistic regression quality is the ROC curve (Receiver Operating Characteristic). The idea is the following.

Consider a very good model like below.



The sensitivity and specificity will be good for sure.

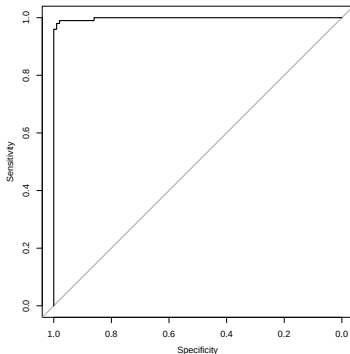
Logistic regression: ROC curve

In fact, these will also be very good, even with a threshold of prediction different from 0.5. In fact, for almost any threshold (btw 0 and 1), the sensitivity and the specificity will be good. This is a sign that the two categories 0 and 1 are very well separated by the model.

The ROC curve represent the sensitivity (y -axis) versus the specificity (x -axis; inverted).

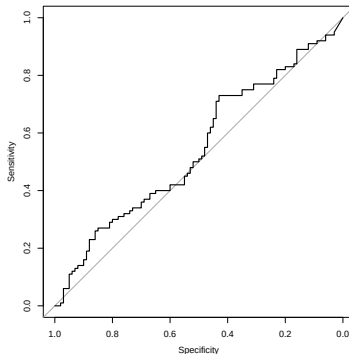
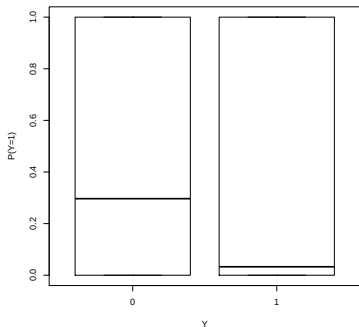
Logistic regression: ROC curve

A good model achieves both large sensitivity and specificity at all threshold levels (like below).



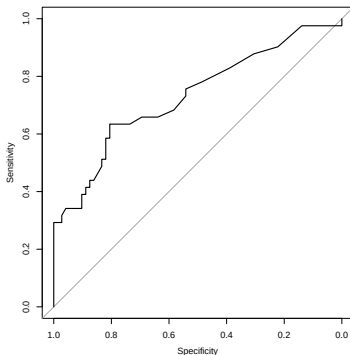
Logistic regression: ROC curve

A bad model (random prediction) achieves a ROC curve close to the diagonal.



Logistic regression: ROC curve

An average model achieves a ROC curve in between the upper corner and the diagonal.



Logistic regression: AUC

AUC stands for "Area Under the Curve". Since the ROC curve identifies good model when they are closer to the left upper corner, one can also compute the area under the ROC curve. The closest to 1 the better the model. An AUC of 0.5 is bad (ROC is diagonal).

For the 3 previous examples:

- ▶ Good: 0.998
- ▶ Bad: 0.538
- ▶ Avg: 0.731